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10 Quant Projects to make your Resume Attractive

1. Options Pricing and Greeks Analysis

- Priced European, American, and exotic options using Black-Scholes, binomial/trinomial trees, and Monte Carlo simulations.
- Calculated and analyzed Greeks (Delta, Gamma, Vega, Theta, Rho) to assess options' sensitivity to market variables and improve hedging strategies.
- Applied the models to price exotic options such as Asian and barrier options, exploring their behavior under varying volatility scenarios.

2. Yield Curve Construction and Analysis

- Constructed yield curves using Nelson-Siegel, Nelson-Siegel-Svensson, and cubic spline interpolation methods to model the term structure of interest rates.
- Analyzed the impact of macroeconomic variables like inflation and monetary policy changes on the shape of the yield curve.
- Extended analysis to compute zero-coupon yields, forward rates, and durations for bond portfolios to aid in risk management and pricing decisions.

3. Monte Carlo Simulation for Derivative Pricing

- Developed Monte Carlo simulation models to price path-dependent derivatives, incorporating advanced stochastic processes like jump diffusion.
- Implemented variance reduction techniques to improve simulation efficiency and convergence for accurate pricing results.
- Applied the model to evaluate exotic derivatives, such as Asian and barrier options, under different market conditions.

4. Portfolio Optimization Using Modern Portfolio Theory

- Developed a portfolio optimization framework to identify optimal asset weights, maximizing returns for specific risk levels using the Markowitz Efficient Frontier.
- Incorporated constraints like sector caps, minimum/maximum allocations, and transaction costs to adapt the model for real-world applications.
- Conducted stress testing and sensitivity analysis under varying market conditions to ensure portfolio robustness and improve risk-adjusted returns.

5. Credit Risk Modeling for Corporate Bonds

- Built a credit risk model to predict default probabilities using historical financial metrics, credit ratings, and macroeconomic indicators.
- Performed extensive feature engineering, including ratio calculations and trend analysis, to enhance model accuracy and reliability.
- Backtested the model on historical datasets, delivering actionable insights for fixed-income investment strategies and creditworthiness assessments.

6. High-Frequency Market Data Analysis

- Analyzed high-frequency tick data to understand order book dynamics, bid-ask spreads, and market liquidity trends.
- Designed and tested an algorithmic trading strategy leveraging insights from market microstructure analysis to optimize trade execution.
- Conducted extensive backtesting using historical tick data to measure strategy profitability, slippage, and resilience under varying market conditions.

7. Volatility Surface Construction and Analysis

- Constructed volatility surfaces by interpolating implied volatilities across different strikes and maturities for equity options.
- Analyzed volatility skew and smile patterns to evaluate market sentiment and pricing anomalies for exotic derivatives.
- Conducted comparative analysis between implied and realized volatility, identifying arbitrage opportunities and forecasting trends.

8. Factor Model for Equity Portfolio Management

- Constructed a multi-factor model to analyze and predict equity portfolio performance, identifying key drivers like value, momentum, and volatility.
- Decomposed portfolio returns into alpha and beta components, optimizing factor exposures to enhance diversification and performance.
- Performed robust backtesting to evaluate the model's predictive accuracy and provided insights for rebalancing strategies.

9. Fixed-Income Portfolio Hedging with Duration and Convexity

- Designed a hedging framework for fixed-income portfolios using duration and convexity analysis to mitigate interest rate risk.
- Identified optimal hedging instruments, including Treasury futures and swaps, to minimize risk exposure and portfolio volatility.
- Conducted scenario testing under various yield curve shifts to evaluate the strategy's robustness and effectiveness.

10. Statistical Arbitrage Strategy for Pairs Trading

- Developed a statistical arbitrage framework for pairs trading by identifying cointegrated stock pairs using historical price data.
- Created a trading strategy with entry/exit rules based on z-score thresholds, ensuring effective signal generation and execution.
- Backtested the strategy to assess profitability and refined it by incorporating risk metrics, transaction costs, and drawdown analysis.